

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are **entered** in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)

2 8 3 9 7 6 8
N N N N N N N

Date of Birth (DD/MM/YYYY)

0 9 / 0 3 / 2 0 0 4
D D M M Y Y Y Y

Mark with an [X] which question you have completed on this sheet.
Cross one box only.

☒ Q1 ☐ Q2 ☐ Q3 ☐ Q4 ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE

Q1.

You may use both sides but only answer **ONE** question per sheet

(a). $f_s = 8 \text{ kHz} \Rightarrow \text{Number of samples } N = f_s \cdot T = 16,000$
 $T = 2 \text{ s}$

Therefore, the frequency bin spacing $\Delta f = \frac{f_s}{N} = 0.5 \text{ Hz}$

To remove frequency $> 4000 \text{ Hz}$,

• for positive frequency: $k \geq \frac{4000 \text{ Hz}}{0.5 \text{ Hz}} = 8,000$
• for negative frequency: $k \leq N - \frac{4000 \text{ Hz}}{0.5 \text{ Hz}} = 8,000$ } $k = 8,000$ need to be manipulated.

Therefore, $Y(k) = \begin{cases} 0 & , k = 8,000 \\ X(k) & , \text{otherwise} \end{cases}$

(b).

(i). $f_s = 8 \text{ kHz}$.

To map the sample indices to time,

$t(n) = \frac{n}{f_s} = \frac{n}{8000} \text{ s}$

Since the length of samples is $N = 2000$,

$t(N) = \frac{2000}{8000} = 0.25 \text{ s}$

i.e. the recording is 0.25 s .

(ii) The frequency bin spacing $\Delta f = \frac{f_s}{N} = 4 \text{ Hz}$.

On the right part of Figure Q1, we can

find out two obvious frequency peak:

$k_1 = 236$, $k_2 = 332$.

$\Rightarrow f_1 = k_1 \cdot \Delta f = 236 \times 4 = 944 \text{ Hz} \approx 941 \text{ Hz}$

$f_2 = k_2 \cdot \Delta f = 332 \times 4 = 1328 \text{ Hz} \approx 1336 \text{ Hz}$

From Table Q1, it can be found the number is 0.

(iii)

What can be obtained :

- which number has possibly been dialed.

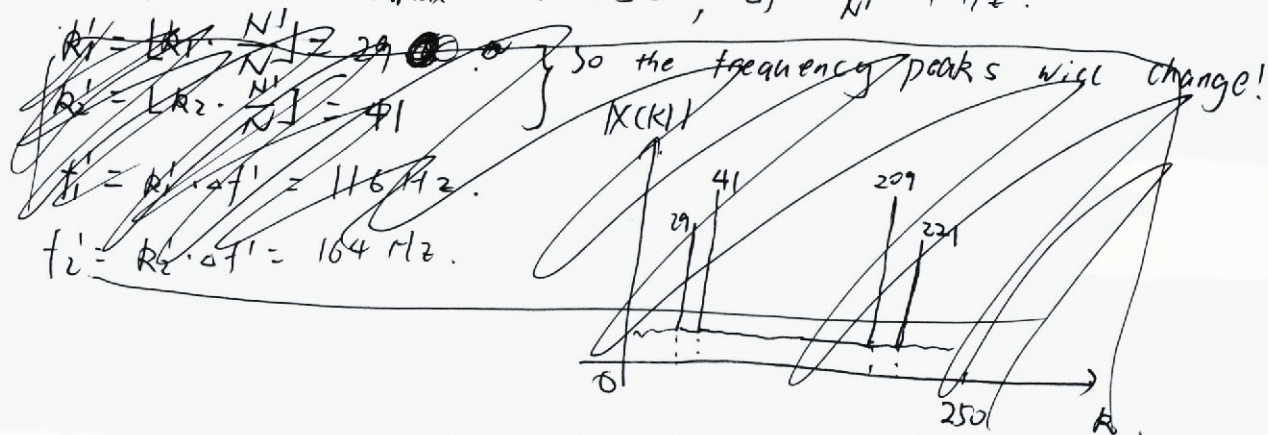
What cannot be obtained :

- the sequence of the dialed numbers.

To fix this :

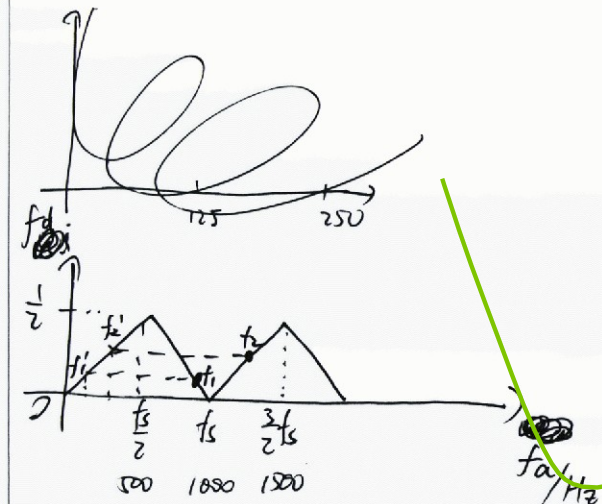
we can separate the single audio into many audio frame in the time domain according to the detected audio magnitude peak. And do FFT for each frame in the same way as ii.

(iv)

If the ~~sample~~ sample time is fixed, i.e. $T = 0.25$ s.then the total sample number is $N' = T \cdot f_s' = 250$ Therefore the $k_{\max} = N' = 250$, $\Delta f' = \frac{f_s'}{N'} = 4$ Hz.

~~$R_1' = \lfloor \frac{f_1'}{\Delta f'} \rfloor = 236$~~
 ~~$R_2' = \lfloor \frac{f_2'}{\Delta f'} \rfloor = 332$~~

So the positions ~~are~~ changed.
 However, the relative position does.

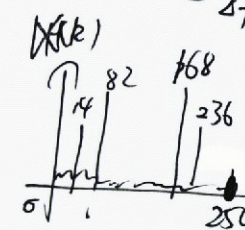


$$f_1' = f_s - f_1 = 56 \text{ Hz.}$$

$$f_2' = f_2 - f_s = 328 \text{ Hz.}$$

$$\Rightarrow R_1' = \lfloor \frac{f_1'}{\Delta f'} \rfloor = 14$$

$$R_2' = \lfloor \frac{f_2'}{\Delta f'} \rfloor = 82$$



k.

5/5